

Black Hole Attack Detection in MANETS

Literature Survey & Work
Plan

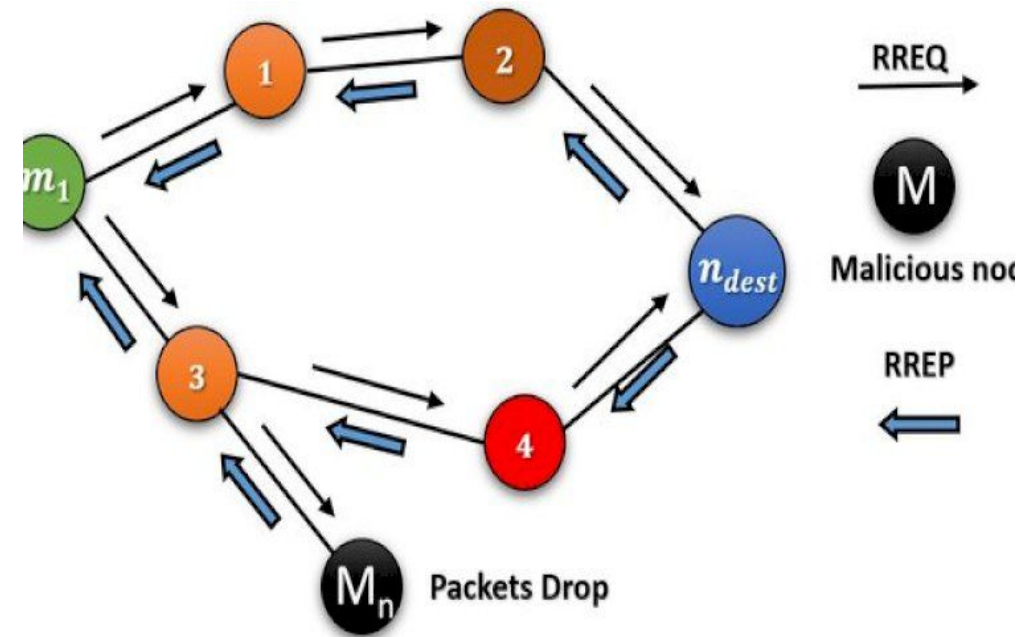
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Problem Statement

Mobile Ad-hoc Networks (MANETs) are decentralized and infrastructure-less, making them highly vulnerable to routing attacks.

- **Black Hole Attack:** A malicious node falsely advertises the shortest path to a destination during the route discovery process.
- **Impact:** Once the route is established, the node silently drops all data packets, causing 100% data loss and network disruption.
- **Challenge:** Detecting these nodes in real-time without compromising network performance is difficult due to the dynamic topology.



Literature Survey & Research Gap

Primary Reference

Abdallah et al. (2024)

International Information & Engineering
Technology Association (IIEETA)

Focus: Proposed the ADS-SVM model (Anomaly
Detection System using Support Vector
Machines).

Technique: Utilized a Radial Basis Function
(RBF) Kernel to handle non-linear data patterns.

Key Result: Achieved **99.96% detection
accuracy** in controlled simulations.

Identified Gap Analysis

While theoretical models achieve high
accuracy, current research lacks practical
implementation tools.

Simulation vs. Reality: Most studies rely on
NS-2/OMNET++ logs rather than live traffic
capture.

Modern Integration: Lack of Python-based,
lightweight implementations suitable for modern
dashboards.

Visualization: Absence of real-time visual
feedback for immediate threat assessment.

Proposed Methodology

1. Detection Engine (Backend)

We will implement the **Support Vector Machine (SVM)** algorithm with an RBF kernel, as recommended by Abdallah et al.
Features: Packet Delivery Ratio (PDR) and Throughput.

Training: Using the NSL-KDD dataset and simulated MANET logs.

2. Real-Time Monitor (Frontend)

A live detection system bridging the gap between theory and practice.

Capture: Python **Scapy** library to sniff live network packets.

Visualization: **Streamlit** dashboard to display network health and flag malicious IPs instantly.

Project Timeline & Work Plan

